

The Coupled Co-Scaling Law

A falsifiable threshold criterion for the stability of recursive self-improvement: correction must out-scale drift-acceleration.

Author: Michael Darius Eastwood (independent researcher; ARC/Eden research programme) · **Working paper**

dated: 26 June 2026 · **Deposited to OSF:** 29 June 2026

OSF: doi.org/10.17605/OSF.IO/6C5XB · **Code:** github.com/MichaelDariusEastwood/arc-principle-validation

OVERVIEW

This project archives the working paper *The Coupled Co-Scaling Law* together with its complete computational apparatus: a runnable verification harness, independent theorem test suites, a pre-registered real-model experiment protocol with its harness, and a SHA-256 integrity manifest.

Most work on recursive self-improvement treats it as a problem of speed: capability may grow explosively and outrun supervision, so the proposed lever is to cap the growth rate. From a minimal dynamical model, this paper derives that the operative lever is different. Stability is set not by the growth rate but by a single inequality between two scaling exponents: the rate at which a misalignment-correcting force strengthens with capability (β) must exceed the rate at which drift accelerates with capability (k). The criterion is $\beta > k$.

THE CLAIM, PRECISELY

Correction strength scales as capability ^{β} and drift acceleration as capability ^{k} , and the system is alignment-stable if and only if $\beta > k$. Its sharpest consequence addresses the central fear of the field: a hard takeoff, even a genuine finite-time intelligence explosion, drives the modelled misalignment fraction to zero if and only if $\beta > k$, and the speed of the explosion does not change that asymptotic verdict. The criterion shares the threshold form of the quantum error-correction sub-threshold condition, offered as a falsifiable hypothesis (the model's suppression law is power-law, not exponential).

WHAT IS ORIGINAL (PRIORITY)

1. the dimensionless control parameter $\rho = \gamma_1 r / A$, the instantaneous drift-to-correction ratio;
2. the sharpened stability criterion $\beta > k$ under accelerating self-improvement;
3. the Hard-Takeoff Depth-Regularity Theorem;
4. identification of the compounding drift channel as the locus of genuine divergence.

The underlying conceptual thesis was first set out in the book *Infinite Architects* (copyright deposited 8 December 2024; published 2 January 2026); the formal, measurable results here are the 2026 form of that thesis.

WHAT IS NOT CLAIMED (HONESTY)

- Not a proof of AI alignment.
- Not an empirically confirmed law: the decisive test, whether real self-improving systems satisfy the criterion, is the open problem. A pre-registered protocol and a runnable real-model harness are provided

to make that test possible.

- The quantum-error-correction correspondence is offered as a falsifiable hypothesis (power-law suppression, not exponential).

STANDING

The closed-form predictions are checked by a runnable verification harness; the theorem statements are independently re-derived in a standalone test suite; and the paper has been hardened against three independent adversarial reviews.

HOW TO CITE

Eastwood, M. D. (2026). *The Coupled Co-Scaling Law: A Falsifiable Threshold Criterion for the Stability of Recursive Self-Improvement*. ARC/Eden research programme. OSF. <https://doi.org/10.17605/OSF.IO/6C5XB>

Priority evidence: book copyright deposited 8 December 2024; *Infinite Architects* published 2 January 2026 (ISBN 978-1806056200); working paper dated 26 June 2026; GitHub commit history; OSF deposit 29 June 2026. The document's 26 June 2026 date is its priority timestamp; the 29 June 2026 OSF date records the deposit.